Dual-Contrastive Sparse Autoencoders Reveal Features of Musical Interpretation

Jan Chen [1 2] **Manuel Cherep** [2] **Pattie Maes** [2] **Nikhil Singh** [3]

Abstract

For decades, philosophers and musicologists have debated which features of a performance are constitutive of the work and which express how it is being interpreted. Computational evidence has been hard to assemble: audio embedding models conflate work identity with performance style, and existing interpretability tools for music generators recover flat dictionaries that mix the two. We address this gap with the dual-contrastive sparse autoencoder (DC-SAE), a two-branch sparse autoencoder that uses coarse work-level metadata to factor a frozen MusicGen transformer's residual stream into work-identity and performance-variation subspaces. Across classical-work, jazz-standard, and pop-cover corpora, the resulting decomposition is supported by probes and feature galleries that surface musically interpretable concepts on each side. Without performer supervision, the variation branch acquires structure related to performer identity, and steering along these directions provides a proof of concept for moving generations toward a target performer's realization while preserving the underlying work. Together, these results show that MusicGen, a generative music transformer, contains a model-internal representation of musical interpretation that can be recovered, interpreted, and steered.

1. Introduction

> *"Performing a Bach fugue with a choir of kazoos may, of itself (although not necessarily), make it a very bad performance; of that there can be no possible doubt. But it cannot, of itself, make the performance a performance of something else."*
>
> – P. Kivy, *Orchestrating Platonism* (Kivy, 1988)

Musical works typically endure by being interpreted, but such interpretations are rarely neutral reproductions of a fully-specified underlying object. Rather, they are representations that differ (sometimes radically) in musical features: minimally, attributes like timing, articulation, ornamentation, timbre, and phrasing. Listeners and practitioners nonetheless treat many of these differences as identity-preserving, i.e. still "the same piece," while treating others as identity-breaking (e.g. resulting in derivative or distinct works). While these judgments are contested, they nevertheless exhibit a structure in the form of stable expectations about which dimensions of variation are constitutive, which are expressive, which are errors, and which alter the work's identity. What has been lacking is an empirical, falsifiable method for characterizing this boundary structure at scale.

Generative music transformers offer such a method, but only obliquely. Models such as MusicGen are trained to model recorded music at scale (Copet et al., 2023), making their residual streams plausible sites for representations of both stable musical content and the expressive behavior that differentiates interpretations of the same work. We ask whether this hypothesized structure can be *extracted* as a measurable decomposition of internal representations, instead of staying implicit in generation behavior. Sparse autoencoders (SAEs) on transformer residual streams (Bricken et al., 2023; Templeton et al., 2024; Gao et al., 2024; Huben et al., 2024) surface monosemantic features corresponding to recognizable concepts, and on MusicGen specifically they recover things like genres, instruments, and timbres (Singh et al., 2026). However, these methods produce flat dictionaries in which work identity and performance variation share the same dimensions, leaving the boundary between the two implicit. We propose a way to make it explicit. Specifically, we contribute:

1. A **dual-contrastive sparse autoencoder architecture** for factorizing musical identity and interpretation (§3)
2. **Empirical evidence** that the factorization separates identity from performance variation (§4)
3. **Examples of interpretable features** correlated with

[1]Technical University of Munich [2]Massachusetts Institute of Technology [3]Dartmouth College. Correspondence to: Jan Chen <chenjan@mit.edu>.

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musical interpretive variation (§4, Appendix I)

4. A **proof of concept for generation-time steering** with learned variation directions (§4)

2. Related Work

Sparse autoencoders provide an interpretable dictionary-learning view of transformer residual streams (Bricken et al., 2023; Templeton et al., 2024; Gao et al., 2024; Huben et al., 2024), and recent work applies this approach to MusicGen (Copet et al., 2023), recovering auditable features such as genre, instrumentation, timbre, and texture, and showing that individual features can steer generation (Singh et al., 2026). DC-SAE builds directly on this model-internal feature space, asking whether it can be structured around the distinction between musical work identity and performance variation. Probing studies similarly show that generative music models encode music-theoretic and symbolic structure (Wei et al., 2024; Ma & Xia, 2024; Ma et al., 2024), while expressive-performance research studies timing, dynamics, articulation, timbre, and other realization-level differences between performances (Cancino-Chacón et al., 2018; Lerch et al., 2020; Hu & Widmer, 2023). Cover-song and version-identification systems learn invariances across different recordings of the same musical work or song family, typically for retrieval (Du et al., 2021; Araz et al., 2024). Content/style disentanglement methods also separate stable content from changing realization, but often in settings where the factors are more explicitly defined, such as pitch versus timbre, audio-effects style, speaker identity, or score-grounded piano performance (Koo et al., 2023; Wu et al., 2025; Zeng et al., 2025). We connect these threads by using work-level metadata to factor MusicGen activations into two sparse, inspectable branches, with one aligned to identity-preserving structure and the other to variation across realizations. Appendix A gives additional related work.

3. Method

Datasets We instantiate our method on three cohorts:

- **Classical**: top-100 works from a reviewed 1,500-piece classical seed built from a Wikidata–MusicBrainz hybrid work manifest (Vrandečić & Krötzsch, 2014; MetaBrainz Foundation, 2026) with IMSLP cleanup (International Music Score Library Project, 2026) and manual movement-level curation (101.2 hours)
- **Jazz**: top-100 most-played standards from a 1,460-tune iReal Pro forum thread (iReal Pro Forum Contributors, 2026) (75.6 hours)
- **SHS100K**: top-100 cover-song families from *SHS100K* (Du et al., 2021) (58.7 audio-hours)

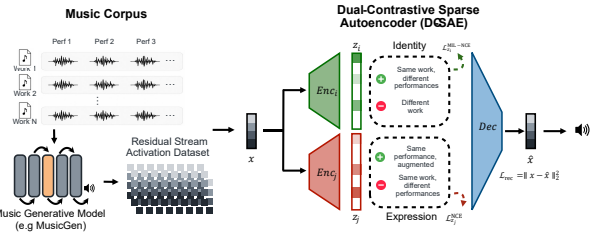


Figure 1. DC-SAE factorizes MusicGen layer-24 activations into a work-identity branch z_i and a performance-variation branch z_j. Same-work pivots pull z_i together while serving as hard negatives for z_j, encouraging the two sparse codes to specialize.

Training uses audio and work identifiers only; performer or artist labels are loaded only for evaluation probes.

Activation extraction We use *MusicGen-Large* (Copet et al., 2023), a 3.3B-parameter autoregressive transformer trained on roughly 20k hours of licensed music. We extract residual-stream activations at layer 24, a middle-depth layer that tends to yield more features (Singh et al., 2026). Each 30-second window yields activations $x \in \mathbb{R}^{1501\times 2048}$.

Pivot windows For each window, our method needs (i) "pivot" windows from *different recordings of the same work* and (ii) distinct-work negatives. We use *MERT-v1-330M* (Li et al., 2024) embeddings to rank all candidate windows whose `work_id` matches the anchor's and retain the top-κ as the same-work pivot bag. These become positives in one objective and negatives in the other, discussed below.

Architecture and training objective The DC-SAE consists of two parallel TopK sparse autoencoders (Makhzani & Frey, 2013; Gao et al., 2024) that share a single linear decoder over concatenated codes (Figure 1). The total loss combines a reconstruction term and two contrastive terms:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{rec}} + \alpha_i \mathcal{L}_{z_i} + \alpha_j \mathcal{L}_{z_j}. \tag{1}$$

For each anchor in a batch we form four views: x_{base}, the clean activation; x_{aug}, the same audio after deterministic acoustic augmentation; a bag of same-work different-recording pivots $\{x_{\text{piv}}^{(1)}, \ldots, x_{\text{piv}}^{(\kappa)}\}$ ranked by MERT cosine similarity (§3); and a bag of different-work negatives. The reconstruction loss is computed only on x_{base}:

$$\mathcal{L}_{\text{rec}} \;=\; \left\|\hat{x}_{\text{base}} - x_{\text{base}}\right\|_2^2. \tag{2}$$

The work-identity term uses multi-positive set NCE (Miech et al., 2020), pooling log-sum-exp similarity over a bag of

same-work pivots:

$$s_\tau(a,u) = \exp\big(\langle a,u\rangle/\tau\big), \tag{3}$$

$$\mathcal{L}_{\text{MIL-NCE}}(a;\,P,\,N;\,\tau) = -\log \frac{\sum_{p\in P} s_\tau(a,p)}{\sum_{q\in P\cup N} s_\tau(a,q)}. \tag{4}$$

The variation term uses the single augmented view as its positive and same-work different-recording pivots as fixed-k negatives:

$$P_i = \{z_i^{\text{piv},1:\kappa_i^+}\}, \qquad N_i = \{z_i^{\text{neg},1:k_i^-}\}, \tag{5}$$

$$\mathcal{L}_{z_i} = \mathcal{L}_{\text{MIL-NCE}}\big(z_i^{\text{base}};\,P_i,\,N_i;\,0.1\big), \tag{6}$$

$$A_j = s_{0.1}(z_j^{\text{base}}, z_j^{\text{aug}}), \tag{7}$$

$$B_j = \sum_{n=1}^{k_j^-} s_{0.1}(z_j^{\text{base}}, z_j^{\text{piv},n}), \tag{8}$$

$$\mathcal{L}_{z_j} = -\log \frac{A_j}{A_j + B_j}. \tag{9}$$

The crucial design choice is the negative bag used in $\mathcal{L}_{z_j}$: same-work different-recording pivots. Its positive view and negative bag share work identity, so the contrast encourages z_j to retain recording- and performance-level information. Symmetrically, $\mathcal{L}_{z_i}$ uses the same-work pivots as the *positive* bag and different-work windows as negatives. Bag-size values (κ_i^+, k_i^-, k_j^-), training schedule, and more implementation details are in Appendices F and L.

4. Results

We present audio demos at storage.googleapis.com/dc-sae-musicgen-listening/index.html, and encourage readers to listen along.

Disentanglement quality We first test whether z_i carries work identity. We fit a raw balanced logistic probe on training recordings and evaluate on extra held-out recordings of the same 50 works per dataset (chance 0.020; Appendix M).

Table 1. Work-classification accuracy on held-out evaluation. z_i, z_j, and $[z_i;\,z_j]$ are the two DC-SAE branches and their concatenation; MERT and CLAP are dense frozen-encoder baselines.

		DC-SAE				
Dataset	SAE	z_i	z_j	$[z_i;\,z_j]$	MERT	CLAP
Classical	0.508	**0.856**	0.440	0.832	0.492	0.048
Jazz	0.128	0.136	0.104	**0.144**	0.076	0.032
SHS100K	0.232	**0.368**	0.176	0.324	0.148	0.048

DC-SAE z_i wins decisively on Classical. On Jazz, concatenation is slightly higher, likely due to the highly flexible way in which standards are realized. On SHS100K, z_i also wins. Reconstruction remains similar to vanilla SAE.

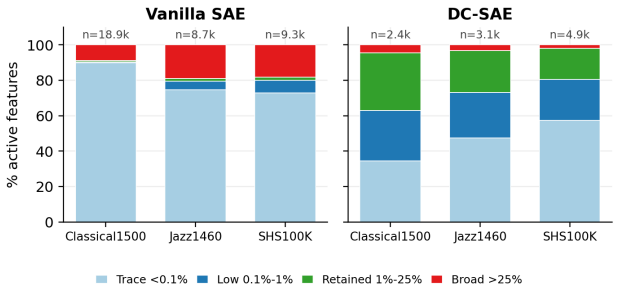


Figure 2. Active-feature prevalence distribution across the three datasets, with dead features excluded.

Feature-prevalence structure DC-SAE also allocates capacity differently from the vanilla SAE. Following Singh et al. (2026), we use 1–25% active prevalence as a practical audit band. In that filtering setup, features below this range are often too obscure to interpret reliably, and highly ubiquitous features tend to be diffuse. Figure 2 asks where each dictionary places its active features under this shared heuristic. The extra mass in the 1–5% bucket is especially useful because it marks selective features with enough examples to audit, and retained-band counts are 4.85 to 5.65× higher for the DC-SAE than the vanilla baseline across the three datasets.

Tradition-specific variation The audited z_j features vary with each corpus's identity criteria. Classical features often track phrasing, rubato, pedaling, and expressive contour; Jazz features pick out lead instrument, ensemble feel, register, and articulation; SHS100K features name accompaniment, vocal texture, orchestration, and production style. The pattern matches thicker and thinner notions of work identity across score-bound works, standards, and covers. Appendix I lists more examples of audited interpretative features.

Robustness and emergent performer structure We next test whether z_j carries same-recording and performer signal. The full clean/augmented table in Appendix M shows that dense encoders carry strong recording-specific evidence: MERT is best before augmentation on the same-recording probe. Because such probes can be solved through recording-channel artifacts, we apply a held-out nuisance recipe not used in DC-SAE training: codec roundtrip, EQ and filtering, colored noise, low-frequency hum, and gain shifts. Table 2 reports this augmented setting.

Under augmentation, MERT and CLAP fall to chance on the same-recording probe, while DC-SAE z_j remains above chance and wins every row. This suggests that z_j retains information about the particular realization that survives a new recording-condition shift.

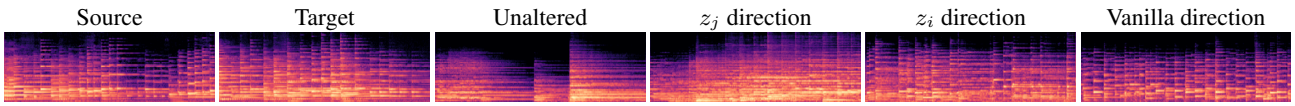


Figure 3. Percentage of aligned-pair contexts in which each rank-1 feature direction beats its matched-random control on MERT target-margin gain; residual is a non-interpretable reference.

Table 2. Augmented same-recording and cross-work performer probe accuracy. Same-recording chance is 0.500; cross-work performer chance is 0.013–0.026.

		DC-SAE				
Dataset	SAE	z_i	z_j	$[z_i; z_j]$	MERT	CLAP
Same-recording: same vs. different recording, same work						
Classical	0.533	0.546	**0.670**	0.631	0.501	0.500
Jazz	0.551	0.635	**0.712**	0.678	0.513	0.500
SHS100K	0.516	0.621	**0.638**	0.613	0.502	0.500
Cross-work performer: performer across unseen works						
Classical	0.178	0.080	**0.218**	0.213	0.132	0.063
Jazz	0.222	0.150	0.257	**0.259**	0.220	0.043
SHS100K	0.209	0.199	0.253	**0.322**	0.229	0.051

The performer rows add a second check because performer labels are never used in training. Under augmentation, z_j or $[z_i; z_j]$ beats the reported MERT/CLAP baselines *on all three datasets*. The Classical row is strongest in z_j alone. The concatenated representation is strongest on Jazz and SHS100K, where performer labels can co-vary with repertoire, arrangement, and instrumentation across works. We interpret this as evidence for emergent performer-related structure, with performer identity still entangled with repertoire and recording context.

Generation-time feature interventions The intervention test asks whether a single sparse feature can move a generation toward a target realization of the same work. We choose same-work source–target windows with high reference MERT similarity as a proxy for shared musical content, then rank features by the target-over-source first-10 s activation contrast. This asks which feature best explains the residual difference between two aligned interpretations. We inject the top feature's decoder direction into MusicGen layer 24 during a 10 s continuation from the source prompt, and score target-margin gain against the matched unaltered baseline, with additional Jazz and SHS100K examples in Appendix K.

Rank-1 z_j steering shifts generations toward the target more often than the matched-random control, on 70.0% of paired contexts overall and on every dataset (75.0% Classical, 62.5% Jazz, 68.8% SHS100K; Table 3). The absolute effects are small, so we present single-feature steering as a proof of concept. Because the aligned pairs are selected by reference-MERT similarity, the score also measures movement in the embedding family used for selection; the CLAP check (Appendix K) and the audited examples mitigate but do not remove this. Within this scope, a single named z_j feature is the most reliable interpretable direction for moving a continuation toward a target realization, recovering much of the movement available in the raw residual stream.

Table 3. Rank-1 feature-direction interventions: percentage of paired contexts in which each system beats its matched-random control on MERT target-margin gain in the aligned-pair diagnostic; residual is a non-interpretable raw-activation reference.

	Interpretable sparse features			Ref.
Dataset	DC z_j	DC z_i	Vanilla	Residual
Classical	**75.0%**	62.5%	56.2%	*65.6%*
Jazz	**62.5%**	56.2%	50.0%	*81.2%*
SHS100K	**68.8%**	46.9%	56.2%	*78.1%*
Total	**70.0%**	55.0%	55.0%	*73.8%*

Figure 3 shows a Classical case. Rubinstein and Yundi Li play the same passage from Chopin's Ballade No. 1. The Yundi target builds toward the later climax with a louder crescendo and more elastic rubato; the Rubinstein source emphasizes that peak less and drops in energy. Feature `f59100` (*expressivo/rubato piano*) captures the contrast, and steering along it produces a louder final span. Vanilla feature `f39785` (*slow introspective solo piano*) favors a quieter texture, missing the climactic emphasis. The example makes the aggregate steering result audible by showing that z_j isolates an interpretation-level cue that changes how the same passage is realized.

These results frame DC-SAE as an empirical instrument for studying musical interpretation in model representations. Separating stable work-level from expressive realization-level cues makes many-realization corpora inspectable through features that can be recovered, interpreted, and steered.

The evidence remains a model-internal account of how MusicGen organizes these corpora, with work labels carrying tradition-specific assumptions and performer signals entangled with repertoire, instrumentation, and recording conditions. Future work should test other model classes and add expert listening studies.

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A. Additional Related Work

Sparse autoencoders and contrastive structure. SAE-based mechanistic interpretability inherits the sparse-coding and dictionary-learning tradition of Olshausen & Field (1997). Its recent transformer applications recover sparse activation features in language and other foundation models (Bricken et al., 2023; Templeton et al., 2024; Gao et al., 2024; Huben et al., 2024), with architectural variants such as TopK, gated, and JumpReLU SAEs (Makhzani & Frey, 2013; Rajamanoharan et al., 2024a;b). Concurrent contrastive SAE work in language specializes a single dictionary across temporal roles (Bhalla et al., 2026); DC-SAE instead uses two explicit branches and work-level audio pairs, so each branch can be probed and steered separately.

MusicGen interpretability and version identity. Music model interpretability has mainly tested known concepts through probing or representation analysis. SynTheory and related MusicGen studies ask whether internal activations encode concepts such as chords, tempo, and pitch structure (Wei et al., 2024; Ma & Xia, 2024; Ma et al., 2024); SAE-based MusicGen discovery asks which concepts emerge without labels (Singh et al., 2026). Our work keeps the unsupervised feature-discovery spirit but adds a weakly supervised identity/variation split. This also differs from cover-song and version-identification systems, which learn invariance across recordings of the same work for retrieval (Du et al., 2021; Araz et al., 2024); here, the within-work variation discarded by retrieval is the object of analysis.

Expressive performance and content/style disentanglement. Expressive-performance modeling treats performance as a structured transformation of a musical object, often measured through timing, dynamics, articulation, intonation, and timbre against a score or symbolic reference (Cancino-Chacón et al., 2018; Lerch et al., 2020; Hu & Widmer, 2023; Lee et al., 2025). Audio and music disentanglement methods similarly separate content from style using raw signals, codec embeddings, symbolic-score grounding, or variance-invariance assumptions (Koo et al., 2023; Zeng et al., 2025; Wu et al., 2025; Lin et al., 2023; Xie et al., 2024). DC-SAE is complementary: it targets a frozen generative transformer's audio representation and uses work-level metadata to define which same-work differences should be treated as interpretation.

B. Preprocessing pipeline

Each track is downmixed to mono, resampled to 32 kHz, loudness-normalized to -18 LUFS using EBU R128 (European Broadcasting Union, 2011), and trimmed with a PANN-based (Kong et al., 2020) onset/offset detector. Trimmed audio is split into non-overlapping 30-second windows; short final windows are zero-padded.

C. Dataset licensing and audit pipeline

We retain only metadata and derived representations for publication; raw audio is not redistributed. Classical-top100, Jazz-top100, and SHS100K-top100 mix publicly accessible recordings with license-tagged catalog rows. Table 4 summarizes the licensing posture of the main external assets as checked for this submission.

Work metadata is normalized, alias-matched, duplicate-checked, and spot-audited before training. Performer labels are evaluation-only.

D. Augmentations

The z_j positive view is an acoustic perturbation intended to preserve work identity and take-level content. We use three deterministic augmentation slots, each seeded by a BLAKE2b hash of $(\texttt{example_id}, s, \texttt{global_seed})$ so resumed runs and diagnostic re-renders see the same positive view. Exact severity ranges live in the released YAML config; Table 5 records only the operation families needed to interpret the method.

The augmentations perturb low-level recording conditions while aiming to preserve the take, timing microstructure, articulation, and ornamentation. They therefore define the positive pair for z_j; same-work different-recording pivots define its negatives.

Table 4. External assets and license posture. "No raw audio release" means the paper release contains metadata, manifests, derived activations, sparse codes, checkpoints, or generated examples where permitted, but excludes downloaded source recordings.

Asset	Role	License / release posture
MusicGen-Large / AudioCraft	Frozen activation source and generation backbone	MusicGen model weights are released under CC-BY-NC 4.0; the AudioCraft code is released under MIT.
MERT-v1-330M	Same-work pair mining and steering metric	Model card license: CC-BY-NC 4.0.
LAION CLAP HTSAT	Dense baseline and secondary audio-similarity check	Model card license: Apache-2.0.
Music Flamingo	Audio captioning for feature autointerpretation	Model card terms: NVIDIA OneWay Noncommercial License; portions of dataset generation are also subject to the Qwen Research License and OpenAI Terms of Use.
Classical metadata	Classical work and movement curation	Wikidata structured data is CC0; MusicBrainz core data is CC0 and supplementary data is CC BY-NC-SA 3.0. IMSLP pages are used only for work/movement cleanup; no IMSLP score files or raw audio are redistributed.
Jazz metadata	Jazz-standard work list and performer overlay	Public iReal Pro forum material is used as a work-identification seed; the release excludes lead sheets and raw audio, and contains normalized work/performance metadata and derived representations only.
SHS100K metadata	Cover-family metadata and public URL seed	Public GitHub metadata repository for SHS100K; no standalone repository license was identified. We cite the dataset source and do not redistribute raw YouTube audio.
Public audio recordings	Source recordings for all three corpora	Cached locally for research use subject to source availability and terms; no raw source recordings are included in the paper release.

Table 5. Composite augmentation slots used to form z_j positives. All slots are applied before MusicGen inference and preserve the underlying take.

Slot	Operation families
`codec_chain_damage`	Room and microphone response, EQ, bandwidth limiting, additive noise, small temporal jitter, lossy codec roundtrip, dynamic compression, soft clipping.
`bad_upload`	Milder room/microphone coloration, EQ, bandwidth limiting, additive noise, small temporal jitter, dynamic compression.
`hard_composite`	Stronger room/microphone coloration, EQ, bandwidth limiting, additive noise, temporal jitter, dynamic compression, soft clipping.

E. Pair coverage

For each anchor window, we retain the top-16 same-work different-recording windows by MERT cosine similarity. During training, the z_i positive bag uses the top $\kappa_i^+ = 4$ eligible in-batch pivots; anchors without eligible pivots contribute only to reconstruction.

Table 6. Pair coverage per dataset under the top-MERT policy. "Eligible anchors" counts windows that have at least one same-work different-recording candidate in the top-16 list; the remainder contribute only to $\mathcal{L}_{\text{rec}}$.

Dataset	Total anchors	Eligible anchors	Coverage (%)
Classical-top100	10,099	10,099	100.0
Jazz-top100	7,609	7,609	100.0
SHS100K-top100	5,994	5,994	100.0

Different-work negatives are sampled uniformly from windows with a different `work_id`.

F. Training hyperparameters

DC-SAE training uses the same token-level recipe across all three datasets, with a uniform 30 s window hop. Table 7 reports the paper-level settings needed to reproduce the objective, model scale, and sampler contract; run-bookkeeping fields are

omitted.

Table 7. DC-SAE training hyperparameters.

Hyperparameter	Value
Backbone activation	MusicGen-Large layer 24 residual stream
Activation dimension	2048
Window length	30 seconds
Window hop	30 s
Training path	token-level; each 30 s window has 1,501 MusicGen tokens
Token subsampling	100 stratified token positions per sampled window
Window-code pooling for probes	mean-pool post-ReLU token activations, then TopK
Branch dictionary width	32,768 (each branch)
Sparsity (TopK)	$k = 50$ per branch
Decoder shared	yes; $\mathbb{R}^{65,536} \to \mathbb{R}^{2048}$ with mean-initialised bias
Parameter count	67.1M per encoder branch; 134.2M shared decoder; 268.5M total
Optimizer	AdamW (Loshchilov & Hutter, 2019), $\beta_1 = 0.0$, $\beta_2 = 0.999$
Learning rate	1×10^{-4}
Weight decay	0.0 for these `w100` runs
Hierarchical batch shape	$g = 16$ works, $p = 5$ performances/work, $r = 4$ windows/performance
Sampled base tokens per step	32,000 tokens ($16 \times 5 \times 4$ windows, 100 token positions/window)
Total steps	5,000
Burn-in (recon-only)	1,000 steps
Contrastive ramp	1,000 steps linear from 0 to full weight
Final α_i	1.0
Final α_j	1.0
Contrastive temperatures	$\tau_i = \tau_j = 0.1$
z_i positive bag size κ_i^+	4 top-ranked same-work different-recording pivots
z_i negative bag size k_i^-	64 different-work windows
z_j positive source	one deterministic augmented view per anchor, sampled from the 3 slots
z_j negative bag size k_j^-	4 same-work different-recording pivots

G. Compute resources

All reported DC-SAE models are trainable on a single NVIDIA H200 GPU. We used up to eight H200 GPUs concurrently as independent one-GPU jobs for different datasets, sweeps, activation extraction, steering generations, or autointerpretation jobs; the method does not require multi-GPU training for a single model. Table 8 reports the resource class used for the main stages.

H. Release and reproducibility package

The code and data package accompanying this workshop paper is not yet public. We intend to open-source the full code used for the paper together with the Hydra configurations, run scripts, split manifests, dataset metadata and validation outputs, derived activations or sparse codes needed for reproduction, trained DC-SAE checkpoints when compatible with upstream licenses, feature-gallery artifacts, and scripts that regenerate the reported tables and figures, alongside an extended version of this work. The release will not include raw downloaded audio; reproduction instructions will describe how to rebuild local audio caches from source metadata where the original source terms permit it.

I. Feature gallery

The supplementary dashboard contains the full curated gallery: 12 features per branch per dataset, with top activating clips, automated labels, acoustic correlations, and ablation checks. This appendix keeps only the standalone audited z_j features needed to read the main text and intervention examples; bundled steering-only examples are left to the audio dashboard.

Curated z_j features (audited). Table 9 lists the audited variation-branch features cited in the main text and intervention triples, with one top-activating spectrogram per feature and a short human-curated label. The full audio gallery is on the

Table 8. Compute resources used to reproduce the paper experiments. GPU stages use one GPU per job; concurrent H200 jobs reduce turnaround time but are not required by the method.

Stage	Worker class	Notes
Preprocessing and metadata validation	CPU, 8–16 workers, 32–64 GB RAM	Audio download, trimming, metadata normalization, duplicate checks, and validation overlays.
MusicGen activation extraction	1 NVIDIA H200 GPU, 8–16 CPU workers, 48–128 GB RAM, 12 h allocation	Extracts layer-24 residual activations from 30 s windows; jobs can be sharded by dataset or split.
DC-SAE training	1 NVIDIA H200 GPU, 8 CPU workers, 48–128 GB RAM, 12 h allocation	Each dataset model trains for 5,000 token-level AdamW steps with the hyperparameters in Table 7.
Probe, prevalence, and dashboard analytics	CPU, 16–32 workers, 96–128 GB RAM	Computes held-out probes, feature-prevalence summaries, gallery metadata, and paper tables.
Steering and autointerpretation	1 NVIDIA H200 GPU, 8 CPU workers, 128 GB RAM, 12 h allocation	Runs 10 s MusicGen continuations, Music Flamingo captioning, and feature-label support checks.

supplementary dashboard.

The full gallery and audio examples are available on the supplementary dashboard.

J. Automated feature-labelling pipeline

For each retained feature, we caption the top ten activating clips with Music Flamingo (Singh et al., 2026), synthesize a short label with a language model, and score label support against the same ten clips. Across 2,830 retained features, the pipeline reports 54% high-confidence labels, 39% medium, and 8% low. Manual inspection shows that these labels are reliable for genre, instrumentation, and gross mood, but less reliable for articulation, rubato, ornamentation, and other performer-level microstructure. For the main-text case studies, we therefore use the automated labels to rank candidates, then report human-audited labels for the selected features.

K. Generation-time feature interventions

Quantitative feature-intervention results referenced in §4 are reported below. The benchmark keeps the top tercile of curated source-target pairs by source-target reference-MERT similarity before generation: 8 of 24 pairs on Classical, 4 of 12 on Jazz, and 8 of 24 on SHS100K. The datasetwise table uses two corpus-level prompts per dataset, yielding 80 paired contexts.

Generation-time intervention. The intervention is applied during MusicGen continuation. No finished waveform is edited. For each ordered direction $A \to B$, the first 10 s of the source recording is used as the prompt, MusicGen generates 10 s, and the generated 0–10 s crop is scored. During generation we add a unit-norm vector v in MusicGen's 2048-dimensional layer-24 residual space:

$$h^{\text{int}}_{24,t} \;=\; h_{24,t} + \alpha\, c\, v,$$

with $\alpha = 1.0$, implementation setting `steering_form=replace`, and `scale_mode=direction_raw_norm`.

Rank-1 feature directions. For each direction $A \to B$, we score features by the positive target-local first-10s contrast $s_f = z_{B,0:10}[f] - z_{A,0:10}[f]$, take the rank-1 feature f^*, weight its decoder row by the target first-10s local maximum, and normalize the result. We report `dual_zj`, `dual_zi`, vanilla SAE, and a raw residual comparator. The matched-random baseline samples a feature from the same pool, bucket-matched on activation scale and prevalence, and reuses the rank-1 weight and norm.

Metric. The primary metric is target-margin gain on the generated 0–10 s crop:

$$\text{margin_gain}(A \to B) \;=\; (T_{\text{int}} - S_{\text{int}}) - (T_{\text{base}} - S_{\text{base}}),$$

where T and S are MERT cosine similarities of the generated clip to the target B and source A reference crops. Audio-quality gates filter generations that fail duration, silence, clipping, or loudness checks before margin aggregation.

Aligned-pair datasetwise table. Table 10 reports the rank-1 MERT margin gain alongside the matched-random control and the fraction of paired contexts in which each system beats its matched-random twin.

CLAP sanity check. CLAP is secondary because the alignment subset is selected by MERT. On the same aligned-pair benchmark, CLAP places DC-SAE z_j first among reported systems (mean margin gain $+0.00292$; beats random on 68.8% of contexts).

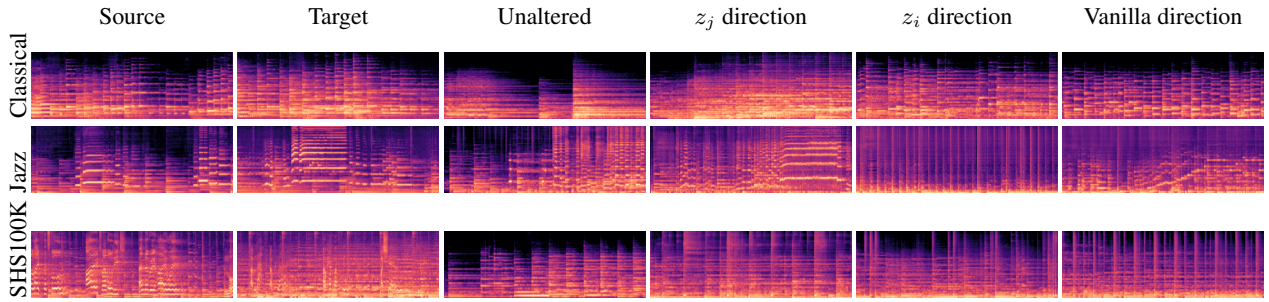


Figure 4. Three audited generation-time intervention triples, one per dataset. Columns show source, target, unaltered continuation, and rank-1 generations produced under z_j, z_i, and vanilla SAE feature directions.

Figure 4 shows the full three-dataset panel. In Classical, the source and target are Arthur Rubinstein and Yundi Li playing Chopin's Ballade No. 1. The z_j feature `f59100` emphasizes the target's crescendo/rubato profile, while the vanilla feature `f39785` favors a slower introspective piano texture. In Jazz, Stan Getz and Art Pepper are both saxophonists, but the target has a sharper, high, sustained tone; z_j feature `f57490` reproduces that high-register cue more directly than the z_i and vanilla directions. In SHS100K, the source is an Andy Williams version with bass and acoustic guitar, while the Tom Jones target has string-ensemble accompaniment; the z_j feature adds a higher, violin-like orchestral line, whereas z_i and vanilla move toward plausible but less target-specific accompaniment.

Audio demos. Paired audio clips for the audited triples are available at `https://storage.googleapis.com/dc-sae-musicgen-listening/index.html`. The page includes source, target, unaltered, z_j, z_i, and vanilla generations for the examples featured in Figure 4.

Vocal-content caveat. In SHS100K examples, the intervention mostly changes accompaniment, texture, register, and articulation, with limited effect on singer identity or intelligible lyric style.

L. Contrastive objectives

Both contrastive objectives use L_2-normalized branch codes and temperature $\tau = 0.1$. The work-identity branch uses multi-positive set NCE (Miech et al., 2020): for anchor a, positive bag P, and negative bag N,

$$\mathcal{L}_{\text{MIL-NCE}}(a;\, P,\, N;\, \tau) \;=\; -\log \frac{\sum_{p\in P} \exp\big(\langle a, p\rangle/\tau\big)}{\sum_{q\in P\cup N} \exp\big(\langle a, q\rangle/\tau\big)},$$

where P_i contains the top four same-work different-recording pivots and N_i contains 64 different-work windows. The performance-variation branch uses a single augmented copy of the anchor as its positive and four same-work different-recording pivots as negatives, implemented as fixed-k InfoNCE. The same pivots therefore pull z_i together and push z_j apart, creating the branch specialization.

M. Probe methodology

Table 11 summarizes the three probes used in the paper. Task A and the performer probe use raw balanced logistic regression on representation means. The same-recording probe uses pair features $[\,|l-r|,\ \langle l, r\rangle,\ \cos(l, r),\ \|l-r\|_2\,]$ followed by `StandardScaler` and balanced logistic regression. Raw ridge, raw `LinearSVC`, and centroid classifiers are retained as artifact-level health checks outside the main paper rows.

The augmented rows in Table 12 apply one deterministic recording-level nuisance recipe per held-out recording: codec roundtrip, EQ, high/low-pass filtering, colored noise, low-frequency hum, and gain shift. This recipe is distinct from the DC-SAE training augmentations, so the augmented rows test held-out recording/channel shift instead of retesting the training positives. Performer labels come from reviewed Classical metadata, an explicit Jazz performer overlay, and an SHS100K performance-artist proxy; they are never used in training.

N. Broader impacts and responsible release

The positive use case is interpretability: DC-SAE gives researchers a way to inspect how a generative music model separates work identity from performance variation, and it can support musicological audits of genre-, tradition-, and performer-sensitive representation learning. The same capability has possible negative uses. Feature directions that encode performer, timbre, arrangement, or production cues could make style imitation easier, and model-internal decompositions could be overread as authoritative judgments about musical ontology or cultural identity. We mitigate these risks by presenting the model as an empirical lens, not an authority; by excluding raw source audio from the release; by documenting the limits of performer and work metadata; and by releasing artifacts for research reproducibility, not as a deployed music-generation product.

Table 9. Audited z_j feature gallery, grouped by dataset. Each row shows a top-activating spectrogram and a human-curated label.

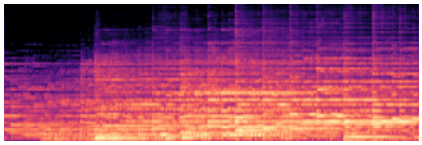
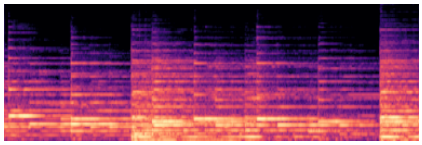
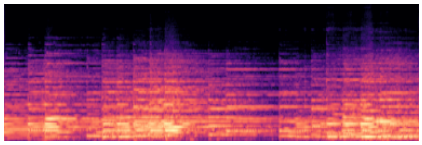
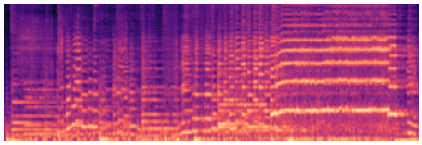
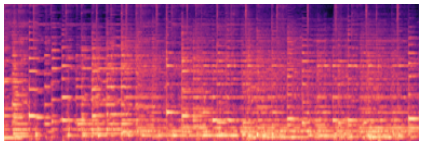
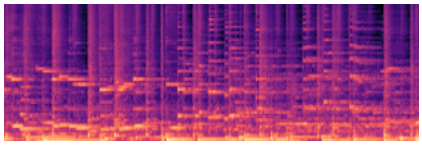
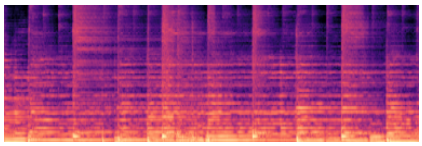
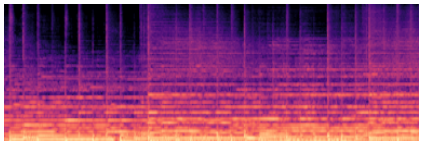
Feature	Top-activating spectrogram	Human label and interpretive role
Classical (Classical)		
`f59100` (featured)		*expressivo / rubato (piano).* Solo piano with expressive phrasing and rubato, Romantic-leaning, especially Chopin.
`f49361`		*sparsely pedaled flowing melody.* Expressive solo piano, sometimes staccato, with flowing melodic line.
`f40006`		*vibrato / wave-like piano.* Wave-like fast arpeggiated chord textures.
Jazz (Jazz)		
`f57490` (featured)		*prominent saxophone / trumpet.* Relaxed slow-tempo jazz with sax or trumpet lead and a sharp higher-pitched tone.
`f41225`		*relaxed-swing saxophone lead.* Saxophone leading a small jazz combo at relaxed swing tempo.
`f3876`		*jazz piano improv.* Fast jazz piano solo lines.
SHS100K (Covers)		
`f48506` (featured)		*cinematic orchestral / choral.* Rich layered textures over operatic and orchestral cover renditions.
`f44666`		*warm emotive ballad vocals.* Vocal ballads with gentle string accompaniment.

Table 10. Aligned-pair MERT margin gain (rank-1, two prompts per dataset, 80 paired contexts). Higher is better. *Lift* is mean margin gain minus mean matched-random gain. *Beats random* is the fraction of paired contexts in which the system exceeds its matched-random control.

Dataset	System	Mean margin gain	Lift vs random	Beats random	Target cosine
Classical	DC-SAE z_j rank-1	$+0.00256$	$+0.00331$	75.0%	0.92652
	DC-SAE z_i rank-1	$+0.00190$	$+0.00265$	62.5%	0.92735
	Vanilla rank-1	$+0.00172$	$+0.00246$	56.2%	0.92733
	Residual	$+0.00151$	$+0.00225$	65.6%	0.93164
	Random matched	-0.00075	0	–	0.92791
Jazz	DC-SAE z_j rank-1	$+0.00472$	$+0.00581$	62.5%	0.81969
	DC-SAE z_i rank-1	-0.00113	-0.00003	56.2%	0.84269
	Vanilla rank-1	$+0.00223$	$+0.00333$	50.0%	0.85489
	Residual	$+0.00500$	$+0.00610$	81.2%	0.84252
	Random matched	-0.00110	0	–	0.83882
SHS100K	DC-SAE z_j rank-1	$+0.00565$	$+0.00444$	68.8%	0.82053
	DC-SAE z_i rank-1	$+0.00243$	$+0.00122$	46.9%	0.82178
	Vanilla rank-1	$+0.00320$	$+0.00199$	56.2%	0.81826
	Residual	$+0.01051$	$+0.00930$	78.1%	0.82363
	Random matched	$+0.00121$	0	–	0.82168

Table 11. Probe protocols used in the paper.

Probe	Split / support	Classifier and metric
Work identity	50 closed-class works per dataset; 5 extra recordings per work held out from SAE training.	Raw balanced logistic regression on recording means; chance $1/50 = 0.020$.
Same recording	Held-out works; balanced same-recording vs different-recording pairs within the same work.	Standardized balanced logistic regression on pair features; chance 0.500; macro-averaged over works and val/test splits.
Performer	Performers with at least three distinct works; grouped work-holdout CV.	Raw balanced logistic regression; performer labels are loaded only at evaluation time from dataset-specific overlays.

Table 12. Full same-recording and cross-work performer probe accuracy under clean and held-out augmented audio. Same-recording chance is 0.500; cross-work performer chance is between 0.013 and 0.026. Bold per-row best, underline second.

	Clean						Aug					
		DC-SAE						DC-SAE				
Dataset	SAE	z_i	z_j	$[z_i;\, z_j]$	MERT	CLAP	SAE	z_i	z_j	$[z_i;\, z_j]$	MERT	CLAP
Same-recording: predict same-recording vs different-recording, given same-work pair												
Classical	0.558	0.546	0.670	<u>0.672</u>	**0.787**	0.599	0.533	0.546	**0.670**	<u>0.631</u>	0.501	0.500
Jazz	0.553	0.649	<u>0.735</u>	0.683	**0.835**	0.600	0.551	0.635	**0.712**	<u>0.678</u>	0.513	0.500
SHS100K	0.531	0.635	<u>0.683</u>	0.675	**0.836**	0.658	0.516	<u>0.621</u>	**0.638**	0.613	0.502	0.500
Cross-work performer: predict performer across unseen works												
Classical	**0.184**	0.080	**0.184**	0.161	<u>0.172</u>	0.069	0.178	0.080	**0.218**	<u>0.213</u>	0.132	0.063
Jazz	0.257	0.176	**0.296**	0.276	<u>0.283</u>	0.052	0.222	0.150	<u>0.257</u>	**0.259**	0.220	0.043
SHS100K	0.247	0.190	<u>0.329</u>	**0.367**	0.260	0.074	0.209	0.199	<u>0.253</u>	**0.322**	0.229	0.051